

Reading the Storm: Media Attribution of United States Climate Disasters, 2000–2024

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Abstract

TODO (150–250 words). Motivation: how local and national newspapers frame the causes of billion-dollar U.S. climate disasters – anthropogenic vs. natural/random attribution – and what drives variation in that framing. Briefly state data (NOAA Billion-Dollar Disasters, 2000–2024, matched to newspaper coverage via Google Custom Search), method (NLP-based Attribution Index at the article and disaster level), and headline findings (to be filled in).

Keywords: climate change attribution, media framing, natural language processing, disaster economics

1 Introduction

TODO: Motivate the question — do U.S. newspapers attribute billion-dollar weather/climate disasters to anthropogenic climate change, or frame them as natural/random variation? Why does this matter (public risk perception, policy salience, insurance markets, disaster preparedness)? Preview data, method, and main results. Outline the rest of the paper.

2 Related Literature

TODO: Position relative to (i) climate communication / media framing literature, (ii) economics of natural disasters and salience, (iii) text-as-data / NLP methods in economics, (iv) political economy of climate policy and local media markets.

3 Data

3.1 NOAA Billion-Dollar Disasters Dataset

The sample is drawn from NOAA’s Weather and Climate Billion-Dollar Disasters dataset, restricted to U.S. events occurring between 2000 and 2024 (313 events). Each event is characterized by a disaster type (Severe Storm, Tropical Cyclone, Flooding, Drought, Wildfire, Winter Storm, or Freeze), a begin and end date, the set of affected states, CPI-adjusted and unadjusted cost (in millions of 2024 dollars), and a fatality count. Severe storms are the most common event type (179 of 313 events), followed by tropical cyclones (48), flooding (30), drought (21), wildfire (19), winter storm (13), and freeze (3). Total CPI-adjusted damages across the sample exceed \$2.36 trillion, with a median event cost of \$2.4 billion (mean \$7.5 billion, driven by a right tail of extreme events up to \$201.3 billion); total reported fatalities across all events are 10,849.

3.2 Newspaper Coverage Data

Newspaper universe. The candidate universe of outlets is drawn from the Access World News (AWN) database, which lists 3,647 U.S. newspapers and wire services with associated city, state, and language metadata. Of these, 970 had no listed website and were excluded. The remaining outlets were passed through an automated verification pipeline that attempted to resolve and HTTP-check each listed URL (classifying each as verified, verified via HTTP-only, or unverified with a status code/reason); 899 outlets ultimately produced a usable, live domain.

Programmable Search Engine (PSE) construction. The 899 verified domains were partitioned into 18 regional Google Programmable Search Engines (PSEs), grouped by the states each outlet primarily serves, plus a 19th “National” PSE covering major national and wire outlets (e.g., the *New York Times*, *Washington Post*, CNN, AP, NPR, USA Today) that are relevant regardless of event geography. Each of the 313 disaster events is mapped to the subset of regional PSEs corresponding to its affected states (plus the National PSE), so that searches are restricted to outlets plausibly covering that event’s geography.

Collection pipeline. For each event, queries combining the disaster name/type and date window are issued against the relevant PSE subset via the Google Custom Search JSON API. Candidate results are filtered on (i) a date window around the event’s begin/end dates (with retrospective coverage allowed up to the start of the next event of the same type, where applicable), (ii) geographic relevance – results must reference a place name consistent with the event’s affected states or broader region, and (iii) disaster-type/keyword relevance. Up to 30 candidate articles are retained per event. The full collection run returned 1,694 candidate articles across the 313 events: 237 events (75.7%) returned at least one candidate article (mean 5.4 articles per event among events with any coverage; 76 events returned none). Of the 1,694 candidate articles, 911 (53.8%) are from outlets on the National PSE list and 783 (46.2%) are from regional/local outlets. These candidate articles are then passed through the relevance-screening and attribution-coding procedure described in Section 4.

3.3 Constructing the Attribution Index

Each of the 1,694 candidate articles is first screened for *relevance* – whether it substantively discusses the specific NOAA-listed event it was retrieved for (correct disaster, geography, and time window), versus being off-topic, ambiguous (**Borderline**), or unusable (**Trash**). Only articles coded **Relevant** proceed to attribution coding.

Each relevant article is then assigned to one of five categories describing how it frames the causal origin of the event, with an associated numeric score $a_{i,d} \in \{-1, 0, 0.5, 1\}$:

Article-level scores are averaged to a disaster-level *Attribution Index*; the aggregation formula, calibration procedure, and treatment of events with no relevant coverage are described in Section 4.

3.4 Summary Statistics

Table 2 summarizes the collection-stage data described above, together with the results of the relevance-screening and attribution-coding pass described in Section 4. Of the 1,694

Table 1: Attribution Coding Categories

Code	Description	Score
A	Explicit anthropogenic climate-change attribution	+1.0
B	Hedged / contextual climate mention	+0.5
C	No causal attribution (pure event reporting)	0.0
D	Explicit non-climate attribution (natural variability or other human causes)	-1.0
E	Mixed / contested framing	0.0 (flagged)

Notes: Full category definitions, decision rules, and example language are given in the project codebook ([paper/attribution_codebook.md](#)). Category E receives the same numeric score as C but is separately flagged as *contested* when computing disaster-level dispersion.

candidate articles, full text was successfully retrieved for 1,591 (93.9%); of those, 1,252 (78.7%) were coded **Relevant** to the specific event they were retrieved for, with the remainder split between **Borderline** (210, 13.2%) and **Trash** (129, 8.1%). Among Relevant articles, the large majority (937, 74.8%) contain no causal attribution at all (category C, pure event reporting); explicit anthropogenic attribution (A) and hedged/contextual climate mentions (B) together account for 252 articles (20.1%), explicit non-climate attribution (D) for 53 (4.2%), and mixed/contested framing (E) for only 10 (0.8%) – contested coverage is rare both at the article level and at the disaster level (only 6 of the 209 covered disasters, 2.9%, have any Contested article at all). At the disaster level, 209 of the 313 events (66.8%) have at least one Relevant article (mean 6.0, median 4 Relevant articles per covered event); the remaining 104 (33.2%) have no Relevant coverage at all and are analyzed as a separate no-attribution group rather than dropped (Section 4). The mean disaster-level Attribution Index among covered events is 0.028 (median 0.000, SD 0.241, range $[-1, 1]$) – i.e. coverage is, on average, only very mildly tilted toward anthropogenic framing and is dominated in practice by events with little to no causal framing either way, consistent with the category-C share above.

4 Empirical Approach

4.1 Article-Level Classification

Each candidate article is coded in two sequential steps, following a written codebook (reproduced in full in the project repository as [attribution_codebook.md](#)) that is also used as the basis for the LLM coding prompt.

Step 1: Relevance screening. Every candidate article is classified as **Relevant** (substantively discusses the specific NOAA-listed event – correct disaster, geography, and time window), **Borderline** (plausibly related but ambiguous, e.g. a brief wire item or a retrospective mention), or **Trash** (off-topic, wrong event, duplicate, paywall/landing page, dead link, or photo gallery with caption-only text). Only **Relevant** articles proceed to Step 2; **Borderline** and **Trash** articles are retained in the data for transparency but excluded from the index.

Step 2: Attribution coding. Each **Relevant** article is assigned to one of five categories describing how it frames the causal origin of the event (Table 1): (A) explicit anthropogenic attribution (+1.0), (B) hedged or contextual climate mention (+0.5), (C) no causal attribution (0.0), (D) explicit non-climate attribution to natural variability or other human causes (−1.0), and (E) mixed/contested framing (0.0, separately flagged as **Contested**). For category D, coders additionally record whether the non-climate attribution is to natural variability (“just weather”) or to other human causes (e.g. land use, infrastructure), since both score −1 on the climate-vs-natural axis but imply different policy narratives.

Full article text retrieval. All coding—both the human calibration pass and the LLM full-sample pass—is performed on the complete article text rather than search-engine snippets. Full text was retrieved in two stages. First, archived copies of each article were obtained via archive.ph, a web archiving service that captures complete page content prior to paywall overlays. For articles unavailable via archive.ph, full text was retrieved manually via Factiva (Dow Jones). This two-stage retrieval procedure was adopted following a finding during calibration that critical attribution language can appear mid-article and is missed with snippet-only coding. Of the 1,694 candidate articles, full text was ultimately retrieved for 1,591 (93.9%); the remaining 103 are excluded from LLM coding and treated as missing in the Attribution Index, comprising 64 articles for which full text could not be recovered by either method (e.g. dead links, HTTP errors, or pages with no archived snapshot), 36 articles that resolved to photo/video-only gallery pages with no substantive article text, and 3 articles – all from `transcripts.cnn.com` – whose URL resolved to a generic chronological listing of segments rather than the specific transcript segment retrieved for that event.

Calibration and validation. Coding is performed by an LLM following the codebook. The LLM is Claude (model `claude-sonnet-4-6`; Anthropic, Inc.), accessed via the Anthropic Messages API, with the codebook supplied as a system prompt and the full article text as the user message. Validation follows a staged procedure that applies two different

reliability bars: a low gate during calibration to catch codebook problems cheaply, and a higher target for the reliability statistics reported in the paper. First, the researcher hand-codes a stratified random calibration sample of 50 articles (stratified by disaster type and by pre-/post-2010 period) for both Relevance and Attribution category. Second, the LLM codes the same 50 articles from the codebook alone. Third, human and LLM codes are compared using unweighted Cohen’s κ for the categorical Relevance (3-category) and Attribution (5-category) codes; if agreement is weak ($\kappa < 0.6$, “moderate” or worse on the Landis and Koch (1977) scale), the codebook wording and examples are revised and the comparison is repeated before scaling up. This 0.6 threshold is intentionally low—its purpose is to catch broken category definitions early, not to certify reliability.

The calibration gate was cleared on the initial pass. Unweighted κ for Relevance was 0.81 (“substantial agreement” on the Landis and Koch 1977 scale, exceeding the $\kappa \geq 0.7$ reporting target). For Attribution, both unweighted and linearly weighted κ were 1.00 (perfect agreement), with a mean absolute difference of 0.00 between human- and LLM-assigned numeric scores. The four residual Relevance disagreements were all boundary cases (**Trash** vs. **Borderline**) where the correct classification is genuinely ambiguous; the relevant decision rules have been noted in the codebook for the full-sample pass.

Once this gate was cleared, the LLM coded all remaining articles in the full sample (1,591 articles with retrievable full text; the 50 calibration-sample articles were not re-coded by the full pass and instead carry forward their calibration-stage LLM codes, for consistency of method across the full **Master** sheet).

Finally, an independent post-hoc audit sample is drawn from the fully coded data, excluded from the calibration sample, for a final human spot-check. Because attribution categories D and E are rare in the full corpus (4.2% and 0.8% of Relevant articles respectively), a simple random draw of the target 100 articles would likely contain too few instances of either to support a meaningful reliability statistic for them. The audit sample therefore deliberately oversamples these categories – including all 10 category-E articles and a random 20 of the 53 category-D articles – with the remaining 70 slots filled by a proportionally stratified random draw (by disaster type and pre-/post-2010 period) from the rest of the coded corpus, for a total of 100 articles. Because of this deliberate oversampling, the audit sample is not a simple random sample of the corpus; reliability statistics computed from it should be read as conditional on this design (informative about each category’s reliability individually) rather than as a corpus-representative error rate, and this is noted explicitly when those statistics are reported. The audit sample is hand-coded blind to the LLM’s codes and then compared to them. For Relevance we report unweighted Cohen’s κ (target $\kappa \geq 0.7$); for Attribution, we report both unweighted κ and a weighted κ (linear weights)

that reflects the ordinal structure of the underlying scores $\{-1, 0, 0.5, 1\}$, since unweighted κ treats an A-vs-D disagreement the same as a B-vs-C disagreement despite the very different implied score gap (target $\kappa_w \geq 0.7$, ideally ≥ 0.8); and we additionally report the mean absolute difference between human- and LLM-assigned numeric scores as a directly interpretable companion statistic. Should the audit-sample statistics fall short of these targets, this is disclosed and discussed as a limitation, since the boundaries between categories B/C and C/E are genuinely fuzzy even for human coders.

The audit sample has been completed. For Relevance (all 100 audited articles), raw agreement was 99% and unweighted Cohen’s $\kappa = 0.96$, comfortably exceeding the $\kappa \geq 0.7$ target; the single disagreement was a Trash/Borderline boundary case. For Attribution, restricted to the 86 articles both the human coder and the LLM independently classified as Relevant (the subsample where both raters have a meaningful 5-category code), raw agreement was 95.3%, unweighted $\kappa = 0.93$, linearly weighted $\kappa_w = 0.96$, and quadratically weighted $\kappa_w = 0.98$ – all well above the $\kappa \geq 0.7$ target and the $\kappa_w \geq 0.8$ “ideal” threshold. The mean absolute difference between human- and LLM-assigned numeric scores was 0.023 on the $[-1, 1]$ scale. All four Attribution disagreements fell on adjacent categories on the causal-framing spectrum (two A/B, one B/C, one B/E) – exactly the kind of boundary fuzziness anticipated above – with zero disagreements spanning a larger gap (e.g. no A-vs-D confusions). Full counts and the complete confusion matrices are reproducible from `code/audit_reliability.py`.

4.2 Disaster-Level Attribution Index

For each disaster d with N_d articles coded **Relevant**, article-level scores are averaged to a disaster-level *Attribution Index*:

$$\text{AttributionIndex}_d = \frac{1}{N_d} \sum_{i=1}^{N_d} a_{i,d}, \quad a_{i,d} \in \{-1, 0, 0.5, 1\} \quad (1)$$

where $a_{i,d}$ is the numeric score for article i covering disaster d from Table 1, with -1 indicating explicit non-climate (natural/random or other human-cause) attribution and $+1$ indicating explicit anthropogenic climate-change attribution. Category E (mixed/contested) scores 0, identically to category C (no attribution), but is additionally flagged so that a disaster-level *Contested share* – the fraction of N_d articles coded E – can be reported separately as a measure of framing disagreement, distinct from the simple absence of any causal claim.

For the 76 events with $N_d = 0$ (no relevant articles, including events that returned

zero search results entirely), $\text{AttributionIndex}_d$ is left undefined (missing) rather than imputed, and these events are analyzed as a separate “no media attribution” group rather than dropped from the sample. All of the above quantities – N_d , $\text{AttributionIndex}_d$, and the Contested share – are computed via formulas in the **All Disasters** sheet of the project workbook (`attribution.xlsx`), which references the article-level codes in the **Master** sheet.

4.3 Specification

We relate the Attribution Index to disaster and outlet characteristics at two levels, deliberately. Disaster type, severity, geography, and time are properties of the *event* and do not vary across the articles covering it, so they are tested in a disaster-level cross-section. National/local status and outlet political lean, by contrast, vary *within* a disaster – the same wildfire is often covered by both a wire service and a local paper – so collapsing them to one disaster-level number would discard most of the identifying variation; these are instead tested in an article-level specification with standard errors clustered by disaster.

Disaster-level specification. For each disaster d with $N_d \geq 1$ (i.e. excluding the 104 events with no Relevant coverage):

$$\begin{aligned} \text{AttributionIndex}_d = & \beta_0 + \boldsymbol{\delta}' \text{Type}_d + \beta_1 \ln(\text{Cost}_d) + \beta_2 \ln(\text{Deaths}_d + 1) + \beta_3 \ln(\text{Duration}_d) \\ & + \beta_4 \text{CoastalShare}_d + \beta_5 \text{RedShare}_d + \beta_6 \text{Year}_d + \beta_7 N_d + \varepsilon_d \end{aligned} \quad (2)$$

where Type_d is a vector of disaster-type dummies (reference category: Severe Storm, the modal type); cost and deaths are log-transformed given their strong right skew ($\ln(\text{Deaths}+1)$ to accommodate zero-fatality events); CoastalShare_d and RedShare_d are, respectively, the share of d ’s affected states that border an ocean or the Gulf of Mexico and the share that voted Republican in the 2020 presidential election – shares rather than a single “primary state” because the **States** field lists affected states alphabetically, not by impact, so no reliable primary state can be recovered for multi-state events; and N_d (the number of Relevant articles) is included as a control for coverage volume. Standard errors are heteroskedasticity-robust (HC1).

Article-level specification. For each Relevant article i covering disaster d :

$$a_{i,d} = \gamma_0 + \gamma_1 \text{National}_i + \gamma_2 \text{OutletLean}_i + \boldsymbol{\delta}' \text{Type}_d + \gamma_3 \text{Year}_d + u_{i,d} \quad (3)$$

where National_i is the existing **National Newspaper** indicator and OutletLean_i is the outlet’s Ad Fontes Bias score (continuous, -42 to $+42$, more positive = more right-leaning), joined by domain; standard errors are clustered by disaster d to account for non-independence among articles covering the same event. OutletLean_i is only defined for the 33 of 135 distinct domains in the sample with an individually-rated page on Ad Fontes/AllSides (covering 71% of Relevant articles, since coverage is concentrated in large outlets), so specifications including it are estimated on that subsample.

5 Results

All results below are estimated on the 209 disasters with $N_d \geq 1$ (or the relevant article-level subsample). The independent post-hoc audit described in Section 4 has been completed and clears the reliability targets ($\kappa = 0.96$ for Relevance, weighted $\kappa_w = 0.96$ – 0.98 for Attribution); see Section 6 for the full statistics.

5.1 Descriptive Patterns

Figure 1 plots the disaster-level Attribution Index against year, together with a fitted linear trend. The index is centered near zero (or negative) for most of the sample’s first decade and rises steadily afterward: mean $\text{AttributionIndex}_d$ is -0.105 (SD 0.250) for disasters beginning before 2010, versus $+0.057$ (SD 0.230) from 2010 onward, a difference that is statistically significant (Welch $t = -3.63$, $p < 0.001$). A bivariate regression of the index on year gives a slope of 0.0129 per year ($p < 0.001$, robust SE), i.e. coverage has moved from net-skeptical/natural framing toward modestly net-anthropogenic framing over the sample period, though even at the end of the sample the average disaster’s coverage remains much closer to “no causal claim” (category C) than to confident anthropogenic attribution.

Figure 2 shows the full cross-sectional distribution underlying that trend. The index is exactly 0.0 for 55.5% of covered disasters (consistent with category C, no causal attribution, being the modal article-level code), with 30.6% net-positive (mean tilt toward anthropogenic framing) and 13.9% net-negative; most of the non-zero mass sits close to zero in both directions, and disasters with a strongly net-anthropogenic ($\text{AttributionIndex}_d$ near $+1$) or net-skeptical (near -1) coverage profile are rare. This reinforces the reading from Section 3.4: the year-over-year movement documented above is a shift in the modest tilt of an otherwise near-zero-centered distribution, not a shift between two opposed extremes.

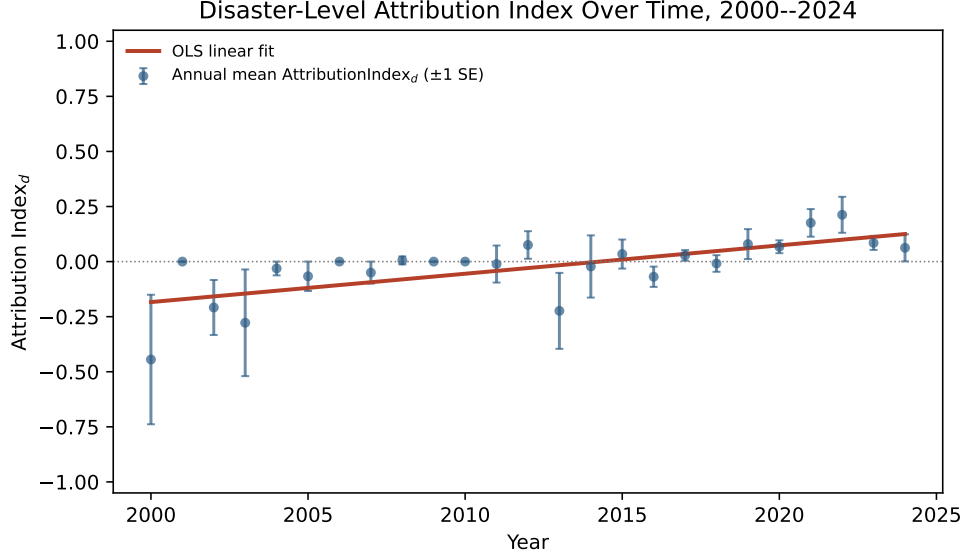


Figure 1: Attribution Index Over Time. Points are annual means across covered disasters (± 1 SE); line is an OLS fit of $\text{AttributionIndex}_d$ on year (slope = 0.0129, $p < 0.001$).

5.2 Variation by Geography

Splitting disasters into “Coastal” (at least one affected state borders an ocean or the Gulf of Mexico) versus “Inland” shows no significant difference: mean Attribution Index is 0.023 (SD 0.235, $N = 179$) for Coastal disasters and 0.070 (SD 0.301, $N = 22$) for Inland disasters (Welch $t = -0.72$, $p = 0.48$). Classifying disasters by the 2020 presidential vote of their affected states (majority-rule across states, for disasters spanning more than one state) likewise shows essentially no difference: 0.023 (SD 0.247) for Majority-Blue disasters ($N = 71$) versus 0.023 (SD 0.239) for Majority-Red disasters ($N = 125$); the 5 Split/Tied disasters average higher (0.213) but the group is too small to draw any conclusion from. A regression of the index on the continuous CoastalShare_d and RedShare_d measures (which use the fraction of a multi-state disaster’s states that are coastal/Republican, rather than collapsing to a single category) confirms both are statistically indistinguishable from zero (Table 5, columns 2–3). Eight disasters with `States = NATIONAL` or limited to Puerto Rico (which has no 2020 presidential vote) are excluded from this section and from any specification including geography.

5.3 Variation by Disaster Type and Severity

Mean Attribution Index varies little across NOAA disaster types – from 0.000 (Freeze, $N = 2$) to 0.074 (Winter Storm, $N = 12$) – and a one-way ANOVA finds no significant difference

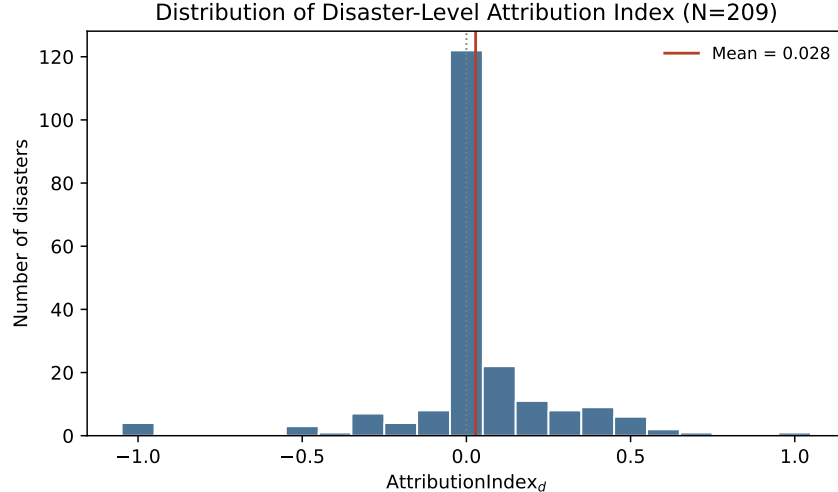


Figure 2: Distribution of the Disaster-Level Attribution Index ($N = 209$ covered disasters). Vertical line at the mean (0.028); dotted line at zero.

across the seven types ($F = 0.32$, $p = 0.93$; full per-type means in Table 3, full type-dummy regression coefficients in Appendix Table 7). Severity, measured as log-CPI-adjusted cost and $\log(\text{deaths}+1)$, is similarly uninformative on its own (correlation with the index of 0.09 and 0.07 respectively, neither coefficient significant at conventional levels in Table 5); event duration (log days) is also not a significant predictor. In short, none of the disaster’s own physical characteristics – what kind of disaster it was, how costly, how deadly, or how long it lasted – predicts how its coverage frames causation, at least in this disaster-level cross-section.

5.4 Variation by Outlet Characteristics

Unlike disaster type, severity, and geography, outlet characteristics show up clearly – but only once a confound is accounted for. In a simple bivariate comparison, articles from national outlets score modestly higher than local outlets (0.120 vs. 0.077 on the $[-1, 1]$ scale; clustered t -test $p = 0.079$), but this gap is not robust to controlling for disaster type and year: in the multivariate article-level specification (Table 4, column 2), the National coefficient falls to 0.031 and is no longer statistically distinguishable from zero ($p = 0.46$). National outlets in this sample skew toward later years and toward disaster types that independently trend toward more anthropogenic framing, and that composition effect appears to be doing the work in the bivariate comparison.

Outlet political lean tells a different story. Using the Ad Fontes Bias score (continuous, more positive = more right-leaning) for the 71% of Relevant articles from an individually-

rated domain, a one-point increase in Bias score is associated with a 0.011-point *decrease* in Attribution Index ($p = 0.013$, clustered SE), controlling for disaster type, year, and national/local status (Table 4, column 2) – i.e. articles from more right-leaning outlets are systematically less likely to frame disasters as anthropogenically caused, even after accounting for which disasters and which years they’re covering. This holds up in the bivariate specification as well (-0.013 , $p = 0.002$) and is robust to using the AllSides score instead of Ad Fontes (-0.033 , $p = 0.003$; Section 6). Figure 3 plots the underlying relationship: a jittered scatter of the four discrete article-level scores against outlet Bias score, with the OLS fit and binned means overlaid – the binned means track the linear fit closely, suggesting the relationship is not an artifact of a nonlinear pattern being forced into a straight line. (Section 6 also shows this effect is concentrated in national-outlet coverage and does not hold among local-only articles.)

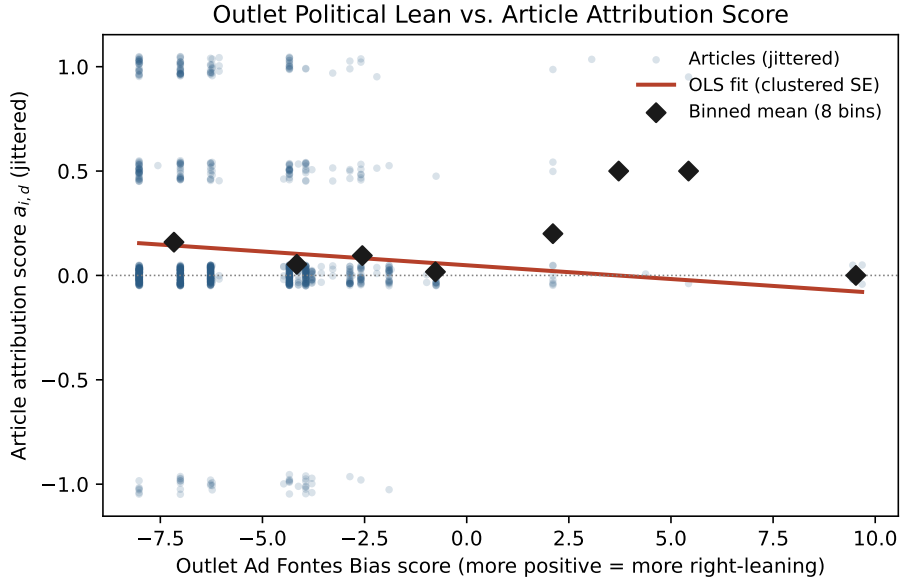


Figure 3: Outlet Political Lean vs. Article Attribution Score ($N = 889$ articles from individually-rated domains). Points are jittered on the y -axis for visibility, since the underlying score only takes four values $(-1, 0, 0.5, 1)$; diamonds are means within 8 equal-width bins of Ad Fontes Bias score.

5.5 Main Regression Results

Table 5 reports the disaster-level specification (equation 2) in three nested columns: (1) disaster type and severity only; (2) adding geography and duration; (3) the full specification, adding year and coverage volume (N_d). Disaster type, cost, deaths, duration, and geography

are not significant in any column. Year is strongly significant once added ($\beta = 0.0128$, $p < 0.001$), consistent with Figure 1, and coverage volume N_d is also positive and significant ($\beta = 0.0089$, $p = 0.036$) – disasters that draw more press coverage skew slightly more anthropogenic on average, though the effect is small and we would caution against a causal reading (more coverage could itself be a consequence of a disaster having a salient climate angle, rather than a cause of one). The full specification explains 20.1% of disaster-level variance in the Attribution Index, almost entirely attributable to the year and N_d terms – the disaster’s own type, cost, death toll, and geography contribute almost nothing on top of those.

6 Robustness

Alternative outlet-lean measure. The outlet political-lean result in Section 5.4 uses Ad Fontes’ numeric Bias score because every one of the 33 individually-rated domains in the sample has a clean numeric value there, whereas a handful of AllSides ratings returned only a category label with no exact meter value (mapped, for this check, to a fixed scale: Left = -5 , Lean Left = -2.5 , Center = 0 , Lean Right = $+2.5$, Right = $+5$). Re-estimating the bivariate article-level specification with the AllSides score in place of Ad Fontes gives the same sign and a comparable magnitude (-0.033 , $p = 0.003$, clustered SE, $N = 930$), so the political-lean finding is not an artifact of which rating service is used.

Disaster fixed effects. The article-level outlet-lean specification in Table 4 controls for disaster type and year but only *clusters* standard errors by disaster, leaving open the possibility that some other disaster-level confound drives the result. Re-estimating with disaster fixed effects – so that National_i and OutletLean_i are identified purely from variation *within* the same disaster (e.g. the AP vs. a local paper covering the same wildfire), implemented via within-disaster demeaning on the 144 disasters with ≥ 2 lean-rated articles – leaves the lean coefficient intact (-0.012 , $p = 0.046$) and the national coefficient modestly larger and now marginally significant (0.075 , $p = 0.099$). This is the most demanding test available in this sample, and the lean result survives it.

Alternative attribution measures. Three checks probe whether the year trend (Section 5) and the outlet-lean effect (Section 5.4) depend on the specific $\{-1, 0, 0.5, 1\}$ scoring of the Attribution Index. First, collapsing to a binary “any climate mention” (A/B) vs. “none” (C/D/E) measure leaves both findings significant (year trend $\beta = 0.0100$, $p < 0.001$; outlet lean $\beta = -0.0177$, $p < 0.001$). Second, recoding category-D articles attributed to

other human causes (rather than natural variability) from -1 to 0 – since that framing is not really climate-skeptical, just non-climate – leaves both findings essentially unchanged (year trend $\beta = 0.0102$, $p < 0.001$; outlet lean $\beta = -0.0135$, $p < 0.001$). Third, dropping the 10 category-E (contested) articles entirely, rather than scoring them 0, again leaves both findings unchanged (year trend $\beta = 0.0130$, $p < 0.001$; outlet lean $\beta = -0.0135$, $p = 0.001$).

Sample restrictions. Restricting the disaster-level sample to the 138 disasters with $N_d \geq 3$ (so the index is not a single article’s score) leaves the year trend significant ($\beta = 0.0112$, $p < 0.001$). Restricting the outlet-lean test to post-2010 articles only – since the pre-2010 era contributes just 37 of 209 covered disasters and could otherwise be driving the result – likewise leaves it significant ($\beta = -0.0149$, $p = 0.001$, $N = 779$).

Placebo checks. If outlet political lean is capturing climate-framing specifically, it should not also predict things that have nothing to do with framing. Two such checks are run. First, lean against article word count: this is *not* a clean null ($\beta = -18.7$ words per Bias-score point, $p = 0.075$) – more right-leaning outlets in this sample write modestly shorter articles, so part of the lean-attribution relationship plausibly reflects a generic length/depth difference across outlets rather than climate-framing decisions specifically, a caveat we flag rather than paper over. Second, lean against the covered disaster’s (NOAA-reported, not outlet-chosen) log-cost and log-deaths: both are clean nulls ($p = 0.685$ and $p = 0.300$ respectively), meaning more right- or left-leaning outlets are not systematically selecting into covering more or less severe disasters – the lean effect is not a selection-into-coverage artifact, even if it may partly reflect a style/depth difference per the word-count check.

National-only vs. local-only subsamples. A different question from Section 5.4 (does national status itself predict the score) is whether the two headline effects are *homogeneous* across national and local coverage, or concentrated in one segment. Re-estimating each headline regression separately by outlet type: the year trend holds in both ($\beta = 0.0174$, $p < 0.001$ among national-outlet articles; $\beta = 0.0107$, $p = 0.007$ among local-outlet articles). The outlet-lean effect does not generalize the same way – it is strongly significant among national-outlet articles ($\beta = -0.0308$, $p = 0.001$, $N = 687$) but statistically indistinguishable from zero among local-outlet articles ($\beta = -0.0025$, $p = 0.69$, $N = 202$). This qualifies the political-lean finding meaningfully: it is robust to every measurement and specification choice tried above, including the demanding disaster- fixed-effects test, but appears to be concentrated in (or driven by) national/wire-service coverage rather than holding uniformly across local outlets as well – though the local subsample is also a third the size of the national

one, so this could partly reflect lower power rather than a true zero effect locally. We report this as an open qualification rather than resolve it, since distinguishing the two would require a larger sample of lean-rated local outlets than is currently available.

Ordered logit/probit (functional-form check). Every result above treats $a_{i,d} \in \{-1, 0, 0.5, 1\}$ as cardinal via OLS – a real assumption, since the gap between “no claim” (C) and “hedged mention” (B) need not be the same size as the gap between B and “explicit attribution” (A). Dropping this assumption entirely and re-estimating both headline relationships as an ordered outcome (collapsing C and E to one level, since both score 0: $D < \{C, E\} < B < A$), via both ordered logit and ordered probit, leaves both findings significant under all four combinations: the year trend (logit $\beta = 0.104$, $p < 0.001$; probit $\beta = 0.055$, $p < 0.001$) and the outlet-lean effect (logit $\beta = -0.085$, $p = 0.004$; probit $\beta = -0.043$, $p = 0.006$). Neither result is an artifact of the cardinal-scaling assumption.¹

Year $\times$ outlet-lean interaction. Has the political-lean gap in framing widened or narrowed over the sample period? Adding an $\text{OutletLean}_i \times \text{Year}_d$ interaction term to the article-level model, the interaction is not statistically significant either alone ($\beta = -0.00028$, $p = 0.61$) or with the full set of controls ($\beta = -0.00030$, $p = 0.66$). The point estimate implies a slightly more negative (i.e. modestly larger-magnitude) lean effect by the end of the sample than the beginning, but this is not distinguishable from no change given the standard error – we find no evidence the ideological gap in attribution framing has systematically widened or narrowed over 2000–2024.

Table 6 summarizes all checks. With the exception of the word-count placebo and the local-only outlet-lean subsample, every alternative specification and sample restriction leaves both headline results intact.

Audit results. All results in Section 5 are estimated from the full-sample LLM coding pass. The independent post-hoc audit sample (100 articles, described in Section 4) has now been hand-coded and compared against the LLM’s codes. Relevance agreement was 99% raw / $\kappa = 0.96$ (target $\kappa \geq 0.7$); Attribution agreement, on the 86 articles both raters independently classified as Relevant, was 95.3% raw, unweighted $\kappa = 0.93$, linearly weighted $\kappa_w = 0.96$, and quadratically weighted $\kappa_w = 0.98$ (targets $\kappa \geq 0.7$, ideally $\kappa_w \geq 0.8$), with a mean absolute score difference of 0.023 on the $[-1, 1]$ scale. All four Attribution disagreements were between adjacent categories (two A/B, one B/C, one B/E); none spanned

¹The ordered-model standard errors here are not clustered by disaster (the `statsmodels` ordered-model implementation does not support cluster-robust SE), so these p -values are an approximate check on sign and significance rather than directly comparable to the clustered-SE results elsewhere in this section.

a larger gap. These confirm, on an independent sample, the high reliability already suggested by the calibration-sample agreement ($\kappa = 0.81$ for Relevance, $\kappa = 1.00$ for Attribution) – the results in Section 5 can be read as confirmed rather than provisional.

7 Discussion

TODO: Interpret findings in light of climate communication, risk perception, and policy implications discussed in the Introduction.

8 Conclusion

TODO: Summarize contribution, limitations, and directions for future work.

References

J. R. Landis and G. G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977.

A Data Appendix

A.1 Data and Code Availability

All data and code underlying this paper are available in the project’s GitHub repository, github.com/Hariaksha/climate-disaster. The central data file – the full **Master** sheet of article-level codes, the **All Disasters** disaster-level table, the **Calibration** and **Audit Sample** sheets, and the **Domain Ratings** outlet-lean table – is `attribution.xlsx`. The Stage 3 regression analysis reported in Sections 5–6 is fully reproducible from `code/Stage3_Regressions.ipynb`, and the attribution-coding codebook is `paper/attribution_codebook.md`. Note that `attribution.xlsx` includes full retrieved article text (**Master** column **Full Article Text**), which is copyrighted newspaper content; this is shared here for replication purposes, but re-publication of that column specifically beyond replication use is not authorized by the original publishers.

A.2 Disaster Event List

The full list of 313 NOAA events, with assigned states/regions and PSE engine coverage, is the **All Disasters** sheet of `attribution.xlsx` (not reproduced in full here; see Section 3 for summary statistics).

A.3 Newspaper Source List

The 19 Programmable Search Engines and their domain coverage are documented in `data/Access World News Database.xlsx`; the resulting outlet-level political-lean ratings (**AllSides**, **Ad Fontes**) used in Section 5.4 are the **Domain Ratings** sheet of `attribution.xlsx`.

A.4 Additional Tables and Figures

The full Stage 3 analysis – dataset construction, all regressions and cross-tabs reported in Section 5, plus the robustness checks and placebo tests in Section 6 – is reproducible

from `code/Stage3_Regressions.ipynb` (see Section A.1 for the full data/code availability statement).

Table 2: Summary Statistics: Sample, Coverage, and Coding Outcomes

	Value
NOAA billion-dollar disaster events (2000–2024)	313
Candidate newspaper articles collected	1,694
Events with ≥ 1 candidate article	237 (75.7%)
Events with no candidate articles	76 (24.3%)
Mean articles per covered event	5.4
Maximum articles for a single event	30
Articles from National PSE outlets	911 (53.8%)
Articles from regional/local PSE outlets	783 (46.2%)
<i>Full-text retrieval and coding outcomes:</i>	
Candidate articles with full text retrieved (“found”)	1,591 (93.9%)
Articles coded Relevant (of found)	1,252 (78.7%)
Articles coded Borderline (of found)	210 (13.2%)
Articles coded Trash (of found)	129 (8.1%)
Category A – explicit anthropogenic (of Relevant)	107 (8.5%)
Category B – hedged/contextual climate mention	145 (11.6%)
Category C – no causal attribution	937 (74.8%)
Category D – explicit non-climate attribution	53 (4.2%)
Category E – mixed/contested	10 (0.8%)
<i>Disaster-level Attribution Index:</i>	
Disasters with ≥ 1 Relevant article ($N_d > 0$)	209 (66.8%)
Disasters with no Relevant coverage ($N_d = 0$)	104 (33.2%)
Mean / median N_d among covered disasters	6.0 / 4
Mean AttributionIndex _d (covered disasters)	0.028
Median / SD AttributionIndex _d	0.000 / 0.241
Disasters with any Contested coverage	6 (2.9% of covered)

Notes: Coverage statistics in the top panel describe the raw candidate-article sample prior to relevance screening. The bottom two panels reflect the full-sample LLM coding pass described in Section 4; reliability statistics from the independent post-hoc audit sample are reported separately once available (Section 4).

Table 3: Mean Attribution Index by Disaster Type

Disaster Type	N	Mean	Median	SD
Severe Storm	89	0.005	0.000	0.236
Tropical Cyclone	45	0.055	0.022	0.113
Flooding	25	0.028	0.000	0.175
Drought	18	0.026	0.017	0.476
Wildfire	18	0.048	0.021	0.315
Winter Storm	12	0.074	0.000	0.148
Freeze	2	0.000	0.000	0.000

Notes: One-way ANOVA across the seven types:
 $F = 0.32$, $p = 0.93$.

Table 4: Outlet Characteristics: Article-Level Results

	(1) Dep. var.: $a_{i,d}$ (article attribution score)	(2)
National outlet	0.0308 (0.0418)	0.0308 (0.0418)
Ad Fontes Bias score		−0.0112** (0.0045)
Year	0.0169*** (0.0024)	0.0177*** (0.0027)
Disaster type dummies	Yes	Yes
Clustered SE (by disaster)	Yes	Yes
N (articles)	1,252	889
Clusters (disasters)	209	194
R^2	0.112	0.124

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses, clustered by disaster. Column (1) is estimated on all 1,252 Relevant articles; column (2) is restricted to the 889 articles (71%) from one of the 33 individually-rated domains, since Ad Fontes lean is undefined otherwise. Full disaster-type dummy coefficients are in Appendix Table 8.

Table 5: Determinants of the Attribution Index (Disaster Level)

	(1)	(2)	(3)
	Dep. var.: AttributionIndex _d		
ln(Cost)	0.0226 (0.0188)	0.0254 (0.0203)	0.0120 (0.0175)
ln(Deaths + 1)	-0.0061 (0.0182)	-0.0057 (0.0191)	0.0045 (0.0166)
ln(Duration)		-0.0203 (0.0214)	0.0012 (0.0213)
CoastalShare _d		-0.0191 (0.0664)	-0.0214 (0.0628)
RedShare _d		0.0338 (0.0553)	0.0164 (0.0549)
Year			0.0128*** (0.0036)
N _d (coverage volume)			0.0089** (0.0042)
Disaster type dummies	Yes	Yes	Yes
N (disasters)	201	201	201
R ²	0.015	0.019	0.201

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Heteroskedasticity-robust (HC1) standard errors in parentheses. Sample restricted to the 201 (of 209) covered disasters with classifiable geography (excludes 8 disasters coded **States** = **NATIONAL** or Puerto Rico only). Reference disaster type is Severe Storm; full type-dummy coefficients are in Appendix Table 7.

Table 6: Robustness Check Summary

Check	Estimate	<i>p</i> -value
<i>Year trend</i>		
Baseline	0.0129/yr	< 0.001
Binary attribution measure	0.0100	< 0.001
D-subtype recoded	0.0102	< 0.001
Contested (E) excluded	0.0130	< 0.001
$N_d \geq 3$ only	0.0112	< 0.001
National-outlet articles only	0.0174	< 0.001
Local-outlet articles only	0.0107	0.007
Ordered logit	0.1044	< 0.001
Ordered probit	0.0546	< 0.001
<i>Outlet political lean (Ad Fontes Bias score)</i>		
Baseline (clustered SE)	−0.0132	0.002
Disaster fixed effects	−0.0115	0.046
Binary attribution measure	−0.0177	< 0.001
D-subtype recoded	−0.0135	0.001
Contested (E) excluded	−0.0135	0.001
Post-2010 only	−0.0149	0.001
National-outlet articles only	−0.0308	0.001
Local-outlet articles only	−0.0025	0.689
Ordered logit	−0.0850	0.004
Ordered probit	−0.0434	0.006
× Year interaction	−0.00028	0.611
<i>Placebo checks (outlet lean predicting non-framing outcomes)</i>		
Word count	−18.69	0.075
log(Cost) of covered disaster	−0.012	0.685
log(Deaths+1) of covered disaster	−0.034	0.300

Notes: Year-trend checks are disaster-level OLS, robust (HC1) SE, except the ordered logit/probit rows, which are article-level (ordinal attribution category regressed on year, non-clustered MLE SE). Outlet-lean and placebo checks are article-level OLS, SE clustered by disaster (or within-disaster-demeaned with no intercept, for the fixed-effects row), except the ordered logit/probit and interaction rows (non-clustered MLE SE and clustered OLS, respectively – see notebook for exact specifications). Full detail and code for every row in `code/Stage3.Regressions.ipynb`, Section K.

Table 7: Full Disaster-Type Coefficients, Disaster-Level Model (Column 3 of Table 5)

Disaster Type (ref. = Severe Storm)	Coef.	SE
Drought	0.0050	(0.1260)
Flooding	0.0195	(0.0488)
Freeze	0.1119	(0.0667)
Tropical Cyclone	0.0106	(0.0556)
Wildfire	0.0407	(0.0903)
Winter Storm	0.0790	(0.0695)

Notes: Robust (HC1) SE. None of the six type dummies is individually significant at $p < 0.10$. Companion coefficients (cost, deaths, duration, geography, year, N_d) are in Table 5, column 3.

Table 8: Full Disaster-Type Coefficients, Article-Level Model (Column 2 of Table 4)

Disaster Type (ref. = Severe Storm)	Coef.	SE
Drought	0.3412***	(0.1054)
Flooding	0.1020**	(0.0518)
Freeze	0.1442**	(0.0708)
Tropical Cyclone	0.1203**	(0.0478)
Wildfire	0.2411***	(0.0776)
Winter Storm	0.0663	(0.0481)

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. SE clustered by disaster. Unlike the disaster-level model, disaster type is strongly significant here: at the article level, with disaster-level confounds (cost, geography, etc.) absorbed differently and N roughly 4–6 $\times$ larger, droughts, wildfires, tropical cyclones, floods, and freezes all score significantly higher than severe storms. Companion coefficients (National, Ad Fontes Bias, Year) are in Table 4, column 2. The contrast with Table 7 (where type is insignificant) reflects the different unit of analysis and sample restriction (889 articles from rated domains vs. 201 disasters) and should not be over-read as a contradiction.